# CM/LM Foundations: Publication-Readiness Audit

Correspondence and Logical Matrices: A Boolean Operator Calculus

Independent AI-assisted mathematical, literature, and reproducibility review

26 September 2026

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# I. Executive verdict

**The central mathematics survives this audit. The current manuscript is not yet journal-submission ready. Its strongest defensible publication form is a focused synthesis or methodological article, not a paper whose case rests on a newly discovered Boolean algebra or a substantial new classification theorem.**

The manuscript contributes an organized symbolic/numeric calculus: a numeric truth matrix is transported into a formula-valued polarity matrix; logical pairing becomes evaluation on agreement bits; valuation, pointwise Boolean composition and input-frame changes fit together. The invertible frame normal form and partition-selector characterization are the most effective explanations of that organization. They remain valid when the reference formulas are dependent or contradictory. The arbitrary-arity results follow from general Boolean identities, rather than from a conjectural extrapolation of binary examples.

No central false theorem was found. The present source has repaired the earlier polarity-mask inconsistency, states the all-true valuation qualification, specifies ordered flattenings, separates output complement from unchanged-outer-operation frame transport, and correctly uses ordinary $\mathbb{F}_2$ rank. The remaining mathematical work is mainly precision: explicitly specify the combined symbolic signed-frame action, expand one compressed higher-arity proof step, and remove a local row/column inconsistency in the STP comparison.

The novelty case is narrower than the amount of notation might suggest. Cheng, Zhao and Xu directly state the numeric truth-vector composition law [S1]. Bricken gives earlier binary matrix and bra–matrix–ket logic [S2]. Boolean-algebra-valued matrix and orthogonal-frame constructions give a close antecedent for the formula frame itself [S3]. The general coherence package is explained by Boolean homomorphisms and pullbacks. This does not make the manuscript useless: a well-designed representation and teaching/specification architecture can be publishable. It does mean that its usefulness must be demonstrated, not inferred from the number of theorem environments.

**The 20,873-check assertion is not reproducible from the supplied package.** Its two-line output survives, but the associated generating program, count definition and execution recipe were not located after recursive archive inspection. A separate checker written for this audit passes **2,022,648 explicitly counted checks**. That new result independently supports the finite claims; it does not establish the provenance of the old run.

The exact executable archive source compiles into a clean 31-page PDF, with no unresolved citations/references or overfull-box warnings. The supplied PDF has the same extracted text page by page. There is no separately mounted file containing the inline chat paste, so this audit does not pretend to have computed a bytewise hash of the paste itself.

**Weighted current-readiness score: 71.45/100.** This is an editorial assessment, not a probability that the mathematics is correct or a prediction of acceptance.

| Release or article form | Determination |
|---|---|
| Public preprint | **READY AFTER MINOR CORRECTIONS**, including correction of the verification assertion and the concrete bibliographic/type issues. |
| Journal submission in current form | **MAJOR REVISION REQUIRED**. |
| New-results mathematical research article | **NOT READY on the evidence established here**; a significant new theorem is not demonstrated. |
| Expository/synthesis article | **Promising and the strongest current route**, after a substantial tightening of the narrative and an end-to-end worked example. |
| Technical/methodological specification | **Plausible after revision**, particularly an exact transport contract and a demonstrated conformance/use case. |

There is **no established P0 mathematical blocker**. The exact unresolved P1 issues are recorded in Section XV. The smallest credible revision is not a new research campaign: repair the evidence and citations, state the transport contract, show one complete calculation, and sharply distinguish the foundations from compiler and phase results.

# II. Artifact and version audit

## Authority and version boundary

The user's inline LaTeX is the presumptive authority. The executable copy examined is the file named `Correspondence_and_Logical_Matrices_Boolean_Operator_Calculus_LM_Centered.tex` in the current-manuscripts subdirectory of the 22 September handoff. It agrees with the supplied paste in title, date, current theorem structure, normal form, selector theorem, corrected conventions, appendices and 20,873-check statement. No material difference was identified. This is a content comparison, not a bytewise equality claim about the chat message.

The audited copy is supplied unchanged as `source/audited_foundations.tex`. It contains **1,588 lines and 97,170 bytes**. Its SHA-256 is:

`86f2b8cb901c0af3f1a9c0a726f26aa65091d137d33bda21aa157c0985b6fb48`

An additional copy in the historical original-release archive has the same hash. The separately uploaded prompt is preserved as `source/requested_audit_prompt.txt`. The input portfolio ZIP has SHA-256:

`727d6b0bd5d7b7dba3645e48e9cefe695981ef84c308f3cff579d8c5d66bd7b0`

All manuscript line references below refer to the unchanged audited source, not to a rewritten draft. The PDF is the fresh compilation of that source. The user's Windows path was not accessed and is not represented as an independently inspected filesystem.

## Recursive inspection and its limits

The outer ZIP contains **393 non-directory members**, including three ZIP files. Recursive inspection encountered **19 ZIP occurrences representing 17 distinct ZIP byte contents**. A deduplicated-content traversal recorded **3,463 member visits**, reading approximately 337 MB of uncompressed member bytes. There are **351 distinct Python source contents** in that traversal. These counts describe archive traversal, not the number of documents whose full arguments were reviewed.

The inventory and targeted search include every recursively encountered unique archive, including levels deeper than the initial extraction. Paths, sizes and hashes are preserved in the inventory JSON files. Repeated archive bytes were not needlessly analyzed as independent evidence. Filename/content searches targeted the verification count, the named checker and relevant foundations material. The 53 search-hit files include incidental numeric data and repeated registers, not 53 independent foundations verification records.

| Collection or artifact | Inspection depth and role |
|---|---|
| Current foundations LaTeX and supplied PDF | Full mathematical reading of the current source; fresh build; all 31 pages visually surveyed; important diagrams/tables inspected at larger scale. Primary object of the audit. |
| `CM_LM_Audit_Astra.md` | Extensive targeted reading of theorem audit, repairs, prior-art, examples, novelty, verification and recommendations. This reviews an earlier source, not the current one. |
| LM-centered revision notes and foundations verification output | Full reading. Useful provenance; neither makes its assertions self-verifying. |
| 26 September foundations `RESEARCH_ASSESSMENT.md` | Full reading and independent comparison with current mathematics and primary literature. Its genre verdict is not adopted merely because it exists. |
| Compiler source/PDF in original release | Actual manuscript inspected, especially definitions, frame/fusion architecture, LM valuation/pairing and experimental-status sections. No complete compiler or benchmark audit is claimed. |
| Compiler R1/R2 and response | Relevant mathematical, scope and editorial sections inspected; R2 read in full. Compiler-only requirements are distinguished from foundations obligations. |
| Publication-program preservation and ownership registers | Targeted searches and reading of foundation ownership, missing-checker and overlap entries. These are secondary assessments. |
| Correctness/model-boundary report | Actual standalone source introductory findings and model/type ledger inspected, together with its README and root claim summary. Its modal theorems and computational campaigns were not re-certified. |

| Collection or artifact | Inspection depth and role |
|---|---|
| Phase handoff and source snapshots | Inventory and selected source/context inspection to identify the added convolution/model boundary. No full phase-paper theorem audit. |
| Structural computation/decomposition direction | Research brief and claim/status records inspected. It is an unfinished direction rather than a completed foundations-supporting experiment. |
| Other nested benchmark, modal and historical data | Inventoried and searched where relevant to recovery. Most algorithms, data rows and unrelated proofs were not substantively reviewed or rerun. |

The current conclusions depend on the actual foundations source, reconstructed arguments, the new finite checker, and the identified primary literature. They do not depend on treating portfolio theorem banks, prior verdicts or historical logs as authoritative. Inspection depth is recorded explicitly in the supplied `inspection_record.md`.

# III. Weighted publication-readiness scorecard

The object scored is the **current manuscript and its originally supplied evidentiary package**. The independent checker created during this audit is available for a revision, but it has not silently repaired the current manuscript's historical claim. Scores are ordinal judgments on the requested 0–10 rubric; the decimal total is arithmetic, not statistical precision.

| Category | Weight | Score | Confidence |
|---|---|---|---|
| Mathematical correctness | 20% | 9.0 | High |
| Proof completeness and assumptions | 10% | 8.5 | High |
| Novelty and priority differentiation | 15% | 4.5 | Medium |
| Significance and substantive contribution | 8% | 6.0 | Medium |
| Literature and bibliography | 8% | 6.5 | Medium |
| Coherence, typing and notation | 7% | 8.5 | High |
| Reproducibility and computation | 7% | 4.0 | High |
| Closure of previous feedback | 6% | 8.0 | Medium |
| Scope and cross-paper differentiation | 6% | 8.0 | Medium |
| Exposition and examples | 5% | 6.5 | High |
| Artifact, build and references | 4% | 9.0 | High |
| Positioning and claim calibration | 4% | 7.5 | Medium |

The weighted score is $\sum_i w_i s_i/10 = \mathbf{71.45}$ out of 100, where the weights sum to 100.

**Mathematical correctness (9.0/10).** Strength: The core identities survive direct proof reconstruction and the finite tests. No false central theorem found. Weakness: No proof-assistant certification; a few local typing and scope clarifications remain. Required action: Add the explicit symbolic transport identity and repair the STP row/column convention.

**Proof completeness and assumptions (8.5/10).** Strength: The general proofs use Boolean identities, partitions and ordinary rank factorization, not extrapolation from small cases. Weakness: Higher-arity pairing is compressed; several convention-dependent statements deserve an extra line. Required action: Expand the selector-to-agreement step; state the empty-XOR convention and the scope of free generators.

**Novelty and priority differentiation (4.5/10).** Strength: Current wording acknowledges the principal numeric antecedents and largely avoids firstness claims. Weakness: The frame matrix has a close explicit Boolean-space antecedent; coherence reduces to preservation by Boolean homomorphisms. No substantial new theorem established. Required action: Position as synthesis/specification and compare the exact mechanisms, especially Gudder–Latremoliere.

**Significance and substantive contribution (6.0/10).** Strength: Typed alignment, invertible symbolic frames and agreement pairing offer a coherent teaching and specification vocabulary. Weakness: A reader still needs a convincing worked task where the notation materially prevents a mistake or simplifies a derivation.

Required action: Add one end-to-end example and a representation/operation comparison, rather than more elementary theorem statements.

**Literature and bibliography (6.5/10).** Strength: Strong primary comparisons are available for truth-vector composition, Boolean frames, NPN and spectral truth representations. Weakness: An unresolved preprint entry and a version-specific Eigenlogic attribution need correction; some historical sources were not accessible in full. Required action: Repair the bibliography and identify the versions actually supporting historical assertions.

**Coherence, typing and notation (8.5/10).** Strength: Matrix product, pointwise operations, pairing, tensor reshape and phase convolution are carefully separated. Weakness: Symbolic signed transport is described more informally than the numeric action; vec changes effective orientation. Required action: Write the signed transport contract explicitly; reserve vec for a column vector throughout.

**Reproducibility and computation (4.0/10).** Strength: The historical output is preserved and the advertised finite mathematical claims can be independently checked. Weakness: The generating source and count definition for 20,873 are not recoverable from this package. Required action: Replace or withdraw the historical reproducibility assertion; publish the new checker with its own count and scope.

**Closure of previous feedback (8.0/10).** Strength: The main earlier mathematical repairs are present in the current source. Weakness: Checker recovery, exact attribution and demonstration of usefulness remain open; not every unrelated portfolio review was substantively reviewed. Required action: Close the remaining foundation-specific issues and avoid importing compiler-only requirements.

**Scope and cross-paper differentiation (8.0/10).** Strength: Performance, cyclic multiplication and modal resources are not premises of the foundations results. Weakness: The compiler still repeats the binary LM, valuation and pairing development; the spectral appendix is peripheral. Required action: Assign foundational ownership explicitly and shorten companion duplication.

**Exposition and examples (6.5/10).** Strength: Definitions and examples are substantially clearer than the historical framing. Weakness: Repeated novelty disclaimers and several elementary theorem environments obscure the central narrative. Required action: Consolidate the claim boundary, use a running example and move reference material behind the main argument.

**Artifact, build and references (9.0/10).** Strength: Fresh two-pass compilation succeeds; 31 pages match the supplied PDF text and no unresolved references or overfull boxes appear. Weakness: There is no bytewise hash of the inline chat paste; one figure label is crowded and the last page is sparse. Required action: Freeze the source as a separately attached file; make small layout repairs and ship a manifest.

**Positioning and claim calibration (7.5/10).** Strength: No new Boolean algebra, physical measurement, compression or speedup is asserted. Weakness: The title and repeated emphasis on first-class operators may still suggest a stronger research contribution than demonstrated. Required action: Lead with a precisely scoped synthesis/methodological contribution, not an implied discovery of operator logic.

# IV. Claim and theorem audit

The following ledger covers all **43 numbered theorem, proposition, corollary, definition, example and substantive remark environments** in the current source. The additional ledger covers **30 groups of substantial unnumbered definitions, displays, historical claims and appendix statements**. These are not 73 claims of novelty. Proofs and adjacent explanatory text were considered with each statement.

Codes: **C** = correct as stated under the declared conventions; **S** = essentially standard or a straightforward specialization; **Q** = local clarification or typing repair recommended; **R** = substantially repetitive. A Q is not automatically a false theorem. No central result received a false/counterexample classification.

## Numbered statements

| Result and source lines | Verdict | Independent reason or qualification |
|---|---|---|
| Proposition 2.1: Unique binary representation (L171-173) | C/S | The four assigned bits give a bijection; there are exactly 16 choices. No mathematical depth beyond the truth-table encoding is asserted. |
| Proposition 2.2: CM selection (L187-193) | C/S | A one-hot row and column leave exactly one summand. The selected coefficient is the connective value. |

| Result and source lines | Verdict | Independent reason or qualification |
|---|---|---|
| Remark 2.3: Why XOR–AND rather than OR–AND? (L198-208) | C | OR and XOR reductions agree only when surviving products are disjoint. The displayed two-term counterexample is correct. |
| Proposition 2.4: CM basis decomposition (L213-219) | C/S | The four matrix units form the standard coefficient-space basis; coefficient linearity does not imply affine dependence on logical inputs. |
| Theorem 3.1: Binary LM frame normal form (L334-346) | C/S | The four disjoint minterms prove the normal form; multiplying by the involutory frames proves the inverse. Arbitrary formula dependence is harmless. |
| Theorem 3.2: LM-to-CM valuation architecture (L361-373) | C | Evaluation of each frame is I or the swap. Joint satisfiability is needed only for the special all-true valuation and is now stated. |
| Theorem 4.1: Logical-pairing identity (L400-407) | C/S | Multiplying an external formula state by its reference frame gives the agreement state. Selection then proves the stated identity. |
| Corollary 4.2: Labelled basis reconstruction (L423-432) | C | Agreement with a labelled reference or its complement is the corresponding constant bit. All four coefficients are recovered. |
| Example 4.3: Pairing an implication LM (L434-446) | C | Both implication substitutions are correct, including the vacuous-antecedent case. |
| Theorem 4.4: Valuation commutes with logical pairing (L450-457) | C/S | Valuation preserves every Boolean term used in the contraction. This is not restricted to independent operands. |
| Theorem 4.5: Partition-of-unity selector characterization (L473-482) | C/S | Orthogonal weights remove cross terms and their sum is one. Conversely, coordinate idempotents force orthogonality and unitality. The F-linearity qualification is important. |
| Theorem 5.1: Aligned LM operator superposition and CM valuation (L516-548) | C/S | Cellwise substitution proves the LM identity; valuation and one-hot selection give the numeric corollaries. The numeric result has direct prior art [S1]. |
| Corollary 5.2: Alignment before operator superposition (L553-563) | C/Q | Correct for transports carrying the stated occurrence into the same semantic assignment frame. Specify the symbolic transport formula, not merely a shape match. |
| Example 5.3: Why the frame condition matters (L565-577) | C | The second implication must be transposed. Its XOR with the first is XOR, whereas unaligned identical tables cancel incorrectly. |
| Example 5.4: LM operators as first-class operands (L579-595) | C/R | The OR/AND/XOR identity is correct. It partly repeats the introductory numerical illustration and can become the running example. |
| Theorem 5.5: Matched-frame kernel composition (L603-616) | C/S | The two copies of the same intermediate frame cancel. The resulting scalar kernel sums parity, not existential reachability. |
| Definition 6.1: Correspondence tensor (L648-655) | C/S | This is the truth function indexed as an order-n Boolean tensor. It stores all 2^n coefficients. |
| Definition 6.2: Logical tensor and LM lift (L657-672) | C | The canonical Boolean term is applied to every polarity tuple. Its codomain is the formula algebra generated by the declared free references. |
| Remark 6.3: The lift as a polarity pullback (L674-681) | C/S | At a valuation the chosen polarity has value x XOR alpha XOR 1. The lift is a pullback, a useful exact reduction of the construction. |
| Theorem 6.4: General valuation theorem (L699-709) | C | The one-triggered axis convention agrees with the valued polarity tuple. The earlier reversed-mask criticism no longer applies. |
| Definition 6.5: Pointwise lift (L716-723) | C | Numeric tensors use g directly; formula tensors use its Boolean term extension. Those two interpretations must stay explicit. |
| Theorem 6.6: LM lift and superposition law (L725-740) | C/S | Every polarity cell preserves substitution into the outer term; all-true evaluation proves injectivity in the free-reference setting. |

| Result and source lines | Verdict | Independent reason or qualification |
|---|---|---|
| Corollary 6.7: The LM image is a Boolean subalgebra (L745-747) | C/S/R | Closure under the Boolean operations plus injectivity yields the isomorphism. This is a repackaging of Theorem 6.6, not another independent novelty. |
| Remark 6.8: Priority boundary for the orbit lift (L749-751) | C | The stated priority boundary is mathematically justified: an equivariant orbit is determined by its identity/reference cell. |
| Theorem 6.9: Polarity-equivariance characterization (L758-767) | C/S | Substitution by a polarity mask changes the cell index by that mask. Conversely the all-positive cell determines the tensor. Extra undeclared parameters are not allowed in the stated codomain. |
| Remark 6.10: Boolean clones and superposition (L774-776) | C | The manuscript explicitly stops short of claiming a full cross-arity clone representation; that restraint is correct. |
| Remark 6.11: Valuation as an intertwining map (L778-790) | C/R | Both intertwining identities follow from the valuation and pointwise laws. The terminology is justified, though the identities repeat prior results. |
| Theorem 6.12: CM/LM coherence under Boolean operator superposition (L792-819) | C/Q | All four claims hold. The fourth uses a genuine partition selector, not distributivity of an arbitrary g. Make the simultaneous reference/index transport precise. |
| Remark 6.13: Output complement is a different symmetry (L824-835) | C | The dual outer function is necessary. Output complement is not an unchanged-g automorphism of arbitrary pointwise superposition. |
| Remark 6.14: Pairing preserves the LM operator calculation (L855-865) | C/R | This is a direct specialization of the coherence theorem to XOR and can be integrated into the running example. |
| Theorem 6.15: Higher-arity pairing and reconstruction (L877-892) | C/Q | The general selector chooses the polarity tuple A at each valuation, producing the agreement tuple. The proof is valid but should write this step explicitly. |
| Definition 7.1: Correspondence-map flattening (L924-932) | C/S | The ordered assembly map resolves noncontiguous and permuted blocks. The flattening is a bijective reindexing, not compression. |
| Theorem 7.2: Rectangular bra–map–ket selection (L949-957) | C/S | The two grouped one-hot states select the unique assembled assignment, even for unequal dimensions. |
| Remark 7.3: Comparison with quantum terminology (L962-964) | C | The distinction between a rectangular linear map and a same-space observable is sound. No physical structure is used in the proof. |
| Remark 7.4: Exact size boundary (L966-968) | C | All flattenings retain 2ˆn entries. No efficiency conclusion follows from the reshape alone. |
| Theorem 8.1: Block-lift theorem (L1000-1013) | C/S | The conditional hypothesis f(a,b)=g(a) Theta h(b) proves both displayed formulas entrywise. It is not a universal one-bit block decomposition. |
| Theorem 8.2: Rank and conjunctive/XOR separability (L1018-1039) | C/S/Q | Ordinary F2 rank equals the minimum number of XOR-summed outer products. Nonzero rank-one factors are unique over F2. State k=0 with the empty XOR for the zero matrix. |
| Example 8.3: A $4 \times 4$ correspondence map (L1046-1069) | C | The two supplied truth vectors and all four rows of the implication lift are correct. |
| Theorem 9.1: Signed-frame action (L1084-1092) | C/S | The numeric pullback and its right-action composition direction are correct. Faithfulness of the entire representation is compatible with nontrivial stabilizers of individual functions. |
| Theorem 9.2: Frame changes commute with pointwise composition (L1096-1104) | C/S/Q | Any common index pullback commutes with cellwise g. The symbolic counterpart needs one explicit formula to eliminate double application of a polarity mask. |
| Proposition 10.1: Top ANF coefficient as correspondence parity (L1141-1148) | C/S | Boolean Mobius inversion gives the full monomial coefficient as the parity of all truth values. The explicit n>=1 restriction is correct. |

| Result and source lines | Verdict | Independent reason or qualification |
|---|---|---|
| Corollary 10.2: Binary nonlinearity parity criterion (L1153-1163) | C/S/R | The binary specialization is correct and is a useful visible invariant, but classical rather than an independent discovery. |
| Proposition 13.1: Core Boolean closure (L1285-1287) | C/Q | Every listed operation stays in the scalar Boolean algebra. This means scalar closure, not closure of the special normalized formula-state family under arbitrary vector addition. |

## Other substantial claims and definitions

| ID and locator | Claim | Assessment |
|---|---|---|
| D01: Sections 1-2; lines 93-169 | Formula/value/operator type distinction and the three compositions | **C**. Logical functions, coefficient arrays and formula-valued arrays have distinct types; the prose does not identify pointwise g with matrix multiplication. |
| D02: Equations (1)-(4); lines 139-195 | XOR glyph, true-first states, CM order, contraction | **C**. The order is (11,10,01,00); the two states are one-hot; rotation of I gives the XOR swap matrix. |
| D03: Equations (6)-(7); lines 224-245 | Transpose, left/right swaps and output complement | **C**. Direct substitution proves the transpose and two swap laws; J XOR C is entrywise complement. |
| D04: Equation (8); lines 247-263 | Support order and Boolean difference | **C/S**. Entrywise containment and B AND NOT A are ordinary support inclusion and set difference, not arithmetic division or remainder. |
| D05: Figure 1 and Section 3.1; lines 265-313 | All-true diagonal arrow and free references | **C/Q**. The caption explicitly conditions the arrow on free references. Keep that condition visible when the diagram is reused. |
| D06: Equations (11)-(12); lines 314-332 | Formula frame symmetry and involution | **C/S**. Diagonal products sum X and NOT X; off-diagonal products vanish. This is a Boolean partition permutation. |
| D07: Equation (17); lines 385-398 | State inner product equals equivalence | **C**. The two agreeing minterms are disjoint and their XOR is XNOR. This is not a positive-definite Hilbert inner product. |
| D08: Section 4.1; lines 464-498 | Partition weights with dependent A and B | **C**. The four minterms always partition the unit even if some are identically zero. Dependence does not invalidate the selector theorem. |
| D09: Equation (31); lines 634-646 | Canonical disjoint-minterm extension | **C/S**. Minterms partition the unit. Evaluation agrees with f, including when substituted formulas are dependent. |
| D10: Equations (35)-(36); lines 683-713 | Higher selection and one-triggered axis reversal | **C**. The numeric selector is one-hot and pi_delta composes by XOR of masks. |
| D11: Section 6.4 and Figure 2; lines 792-853 | Meaning of coherence | **C/S**. The common preservation laws are Boolean-homomorphism identities. They do not assert a new algebraic law for arbitrary non-selector contractions. |
| D12: Figure 3; lines 896-911 | Faithful embedding diagram | **C**. For free reference generators all-true evaluation is a left inverse; this suffices for injectivity. |
| D13: Section 7; lines 919-947 | Ordered assembly, tensor states and dimensions | **C/Q**. The array is 2^p by 2^q. The coordinate-space identification supports the endomorphism wording when p=q; semantically the row and column frames remain labelled. |
| D14: Equations (47)-(48); lines 985-998 | Outer lift and symbolic polarity vectors | **C**. These are rectangular arrays on disjoint blocks, not an ordinary matrix product of two existing CMs. |

| ID and locator | Claim | Assessment |
|---|---|---|
| D15: Section 8 after Theorem 8.2; lines 1041-1044 | Rank algebra and bipartition dependence | **C**. F2 rank is distinct from OR-based Boolean rank and can change when variables move between blocks. |
| D16: Equation (54) and following; lines 1071-1083 | Signed map and right-action convention | **C**. With T_phi C=C composed with phi, T_phi T_psi=T_(psi composed with phi). |
| D17: Section 9.1; lines 1106-1119 | NPN transformations | **C/S**. The four displayed binary operations realize standard input negation, permutation and output negation; NPN is not new [S8,S9]. |
| D18: Equations (57)-(58); lines 1121-1139 | ANF and Mobius transform | **C/S**. The Boolean zeta transform is its own inverse in characteristic two. Truth coefficients are not ANF coefficients [S10]. |
| D19: Section 10 STP bridge; lines 1167-1175 | Row truth vector versus vec | **Q: local dimensional notation**. The intended 2 by 4 structure matrix is correct. Since vec is subsequently a column, define t=vec(C)ˆT here. |
| D20: Equations (60)-(62); lines 1177-1232 | Literal clockwise rotation and coefficient permutation | **C**. The explicit matrix sends (a,b,c,d) to (c,a,d,b); the truth function becomes f(NOT y,x). Four iterations are the identity. |
| D21: Section 11; lines 1234-1247 | Impax support and optional cyclic algebra | **C/Q**. I XOR swap is J. The group-algebra isomorphism is correct after introducing convolution; cyclic ordering should be left to the companion construction. |
| D22: Section 12; lines 1249-1256 | Compiler interpretation and no performance inference | **C/Q**. Frame-correct fusion is sound. Whether it saves work in practice is empirical and explicitly outside this paper. |
| D23: Introduction and Section 13 related work | Historical and comparative assertions | **Mixed bibliographic status**. The core numeric comparisons are supported; the Eigenlogic version wording and unresolved Zhao entry require repair. See Sections VII and X. |
| D24: Discussion and conclusion; lines 1295-1336 | Calculus, future clone structure and operator-valued enrichment | **C/Q**. These are proposals or interpretive claims, not proved research results. Keep matrix-valued enrichment separate from formula-valued operands. |
| D25: Appendix A; lines 1340-1376 | Characteristic polynomial and four spectral examples | **C**. All four spectra and the XOR non-diagonalizability claim are correct for XOR-AND field arithmetic. |
| D26: Appendix A; lines 1377-1384 | Eight idempotents and four symmetric idempotents | **C/S**. The entrywise idempotence equations yield eight solutions; symmetry forces off-diagonal entries zero. The independent enumeration agrees. |
| D27: Appendix A; lines 1386-1408 | Matrix elements, diagonal truth observable and valuation spectrum | **C/Q**. D_Theta is a different 4-dimensional representation; independent row/column reversals need not preserve spectrum. Specify set versus multiplicity convention for Spec if the profile is later developed. |
| D28: Appendix B; lines 1411-1432 | Formula-valued LM basis expansion | **C/S**. Its upper-left cell is the disjoint-minterm formula and the other cells are polarity substitutions. Equivalent to the normal form. |
| D29: Appendices C-D; lines 1434-1471 | Compatibility and notation tables | **C/R**. The identities repeat proved statements. The table is useful reference material; consistent vec orientation is still needed. |
| D30: Appendix E; lines 1473-1474 | Accompanying checker and 20,873 exact checks | **Not reproducible as packaged**. A two-line output survives, but the generating checker, count definition and command were not found. The new independent checks must not be passed off as that historical run. |

# V. Proof completeness and hidden assumptions

## The frame and pairing architecture survives dependent formulas

The scalar ring is the Boolean formula algebra modulo logical equivalence, with XOR as addition and conjunction as multiplication. Thus the identities are equalities of equivalence classes, not claims that two strings of syntax are identical. Formula dependence is permitted: $X$ and $Y$ may share variables, coincide, be complementary, or be constants.

For the displayed frame $\mathsf{S}_X$, the diagonal of $\mathsf{S}_X^2$ is $X \Updownarrow \neg X = 1$, and each off-diagonal entry is the XOR of two copies of $X \wedge \neg X = 0$. Hence the frame is involutory. The upper-left entry of $\mathsf{S}_X[\Theta]\mathsf{S}_Y$ is the four-term disjoint-minterm expansion of $X\Theta Y$; the other entries change the appropriate polarities. This proves Theorem 3.1 for arbitrary formulas. Multiplication by the two inverse frames recovers $[\Theta]$ without a valuation or independence assumption.

This distinction matters for $X = P$, $Y = \neg P$. There is no valuation making both true, but the normal form and inverse are still valid. The current manuscript correctly states this. A criticism that such dependence destroys the LM architecture is therefore wrong for this source.

The pairing theorem follows from

$$\langle A|\mathsf{S}_X = \langle A \Leftrightarrow X|, \qquad \mathsf{S}_Y|B\rangle = |Y \Leftrightarrow B\rangle.$$

The first component of the left identity is precisely the disjoint sum of the two agreement cases; the second component is its complement. One-hot selection, interpreted in the formula algebra, then gives the stated connective of the two agreement bits. The result is a relationship formula. It does not claim that pairing is a probability, an expectation, or a test that both agreements separately hold. For implication, a false antecedent agreement can make the paired result true.

There is a useful consequence worth emphasizing more strongly: **the entire LM still encodes the connective after dependent substitution**. For example, when $X = Y = P$, the reference formula $P \Updownarrow P$ is zero, but its full polarity LM is the swap matrix, not the zero LM. Invertible frame reconstruction distinguishes the two. This is a useful explanatory corollary of the normal form, not an additional historical novelty claim.

## Why the selector theorem proves the full outer-operation law

Write $q_w(T) = \Updownarrow_i w_i T_i$. Linearity gives XOR preservation. Orthogonality gives

$$q_w(T)q_w(U) = \Updownarrow_{i,j} w_i w_j T_i U_j = \Updownarrow_i w_i T_i U_i = q_w(TU).$$

The weights sum to one, so $q_w(1) = 1$; complement follows. XOR, conjunction and the unit generate every Boolean term operation. Conversely, for the coordinate idempotents $e_i$, preservation of $e_i e_j = 0$ forces $w_i w_j = 0$, and preservation of the unit forces their sum to one. The stated converse is properly restricted to the displayed formula-linear maps. It is not a classification of arbitrary Boolean-algebra homomorphisms without that qualification.

For logical pairing, weights $A^{\langle r\rangle}B^{\langle s\rangle}$ are orthogonal even when some are identically zero. This is why dependence of $A$ and $B$ does not break coherence. An arbitrary contraction with nonorthogonal weights does not enjoy the same property; weights $(1, 1)$ already fail unitality under XOR.

## The arbitrary-arity statements have general proofs

At an assignment $x$, the value of $X_i^{\langle\alpha_i\rangle}$ is $x_i \Updownarrow \alpha_i \Updownarrow 1$. Therefore

$$\mathcal{M}_f(x, \alpha) = f(x \Updownarrow \alpha \Updownarrow \mathbf{1}).$$

This single identity verifies the indexing convention, valuation formula, orbit structure and the pointwise superposition law for every finite arity. Setting $x = \mathbf{1}$ recovers the truth tensor in the free-reference algebra. Applying a polarity substitution to the reference variables is equivalent to changing the same index mask. Conversely, an equivariant tensor is recovered from its all-positive cell. These are full arguments for arbitrary $n$, not inferences from the arity-three tests.

For higher pairing, at any valuation the product of the $A_i$-minterms selects exactly the polarity tuple $\alpha = (v(A_i))_i$. Substitution into the preceding identity gives

$$f\big(v(A_1) \Updownarrow v(X_1) \Updownarrow 1, \ldots, v(A_n) \Updownarrow v(X_n) \Updownarrow 1\big),$$

which is $f$ on the agreement bits. Since the equality holds for every propositional valuation, the formulas are equivalent. Inserting this short argument would make Theorem 6.15's proof more transparent than its current compressed reference to the selector theorem.

The intrinsic equivariance characterization has a narrower scope than the binary normal form: it is stated in $\mathcal{F}_{\mathbf{X}}$ with free reference generators. An independent negation of one formal generator need not descend to a quotient imposing, for example, $X_1 = X_2$. Do not export that group-action statement to constrained references without specifying the induced action. This is a scope boundary, not a missing hypothesis in the current theorem.

## An exact symbolic signed-frame contract should be added

The numeric right action is correct as written. What deserves an explicit statement is how the symbolic references and polarity indices transform together. Let $P_\sigma z = (z_{\sigma(i)})_i$ and $\phi(z) = P_\sigma z \Updownarrow \epsilon$. The correct identity is

$$\boxed{\mathcal{L}_{\mathbf{X}}(f \circ \phi)[\alpha] = \mathcal{L}_{P_\sigma \mathbf{X}}(f)[P_\sigma \alpha \Updownarrow \epsilon] = \mathcal{L}_{\phi(\mathbf{X})}(f)[P_\sigma \alpha].}$$

All three expressions have value $f(P_\sigma x \Updownarrow P_\sigma \alpha \Updownarrow \epsilon \Updownarrow \mathbf{1})$. A polarity mask may be placed in the references or the slot indices, but not independently in both. Applying it twice cancels it. For one variable, $f$ the identity, $x = \alpha = 1$ and $\epsilon = 1$, the desired transformed value is zero; negating both the reference and the index incorrectly returns one.

This is a proposed clarification of Theorem 6.12(3) and Section 9, **not a refutation of their common-reindexing identity**. It is especially important if the article is sold as a formal specification. Likewise, a claim that a scalar pairing is invariant under changing coordinates must transport the selector states too; the current preservation-of-outer-operation theorem does not assert invariance under a fixed, untransported selector.

## Matrix multiplication, block construction and rank

Matched intermediate references cancel because $\mathsf{S}_Y^2 = I$. With unmatched references, a residual $\mathsf{S}_Y \mathsf{S}_W$ generally remains. The resulting kernel sums mod-two contributions over the intermediate Boolean index. It is not ordinary existential relational composition and is not pointwise conjunction. The current paper distinguishes these correctly.

The flattening proof is valid for any ordered bipartition, including noncontiguous indices. The dimensions are $2^p \times 2^q$ and the two one-hot tensor states have matching lengths. When $p = q$, identifying both coordinate spaces with $\mathbb{F}_2^{2^p}$ supports the endomorphism wording; the semantic row and column labels should nevertheless remain visible. If empty blocks are allowed, add the conventional empty tensor/state of length one.

The block-lift theorem is conditional. It does not say every function has the form $g(a) \Theta h(b)$ with one Boolean output from each block. A two-bit equality relation $f(a, b) = 1$ exactly when $a = b$ has four distinct rows; a one-bit $g(a)$ can distinguish at most two row patterns. That relation is a useful limitation example.

For rank, every conjunctive term gives an outer product and thus rank at most one. A sum of $k$ such matrices has rank at most $k$. A rank factorization gives exactly $r$ outer products, so the minimum is $r$. Nonzero rank-one factors are unique over $\mathbb{F}_2$ because there is only one nonzero scalar. Zero factors are not unique, and $k = 0$ should be admitted through the empty XOR. The invariant depends on the bipartition: $(X_1 \Leftrightarrow X_2) \wedge (X_3 \Leftrightarrow X_4)$ has rank one across $12 \mid 34$ and rank four across $13 \mid 24$.

## Other proof and notation details

The ANF parity statement follows directly from Boolean Mobius inversion; the current $n \geq 1$ condition is correct. The rotation matrix and its associated input transformation agree entry by entry. The spectral statements are correct for XOR–AND field arithmetic: OR has no eigenvalue in $\mathbb{F}_2$ and XOR has a repeated

eigenvalue one with a one-dimensional eigenspace. The four truth outputs are not the two-dimensional CM spectrum.

Two local corrections are advisable. First, the STP comparison uses $t$ as a row while the later vec convention is a column. Define $t = \mathrm{vec}([\Theta])^T$ there. Second, the phrase restricting formula-matrix multiplication to "this subsection only" is misleading because the same convention is used again in the matched-frame theorem. Neither correction changes a mathematical conclusion.

The closure proposition is scalar closure. It does not mean that arbitrary vector addition preserves the special formula-state family: $|A\rangle \Updownarrow |A\rangle = 0$ is not of the form $(B, \neg B)^T$. The present text does not assert that stronger closure, but a one-sentence warning would prevent an easy misreading.

# VI. Independent computational verification

## Historical result: preserved output, missing generating artifact

The archive contains `LM_Foundations_Verification_Output.txt`, a two-line record reporting 20,873 checks and naming the families covered. Revision notes repeat it. The current manuscript says an independent standard-library checker accompanies the revision. The recursive search did **not** locate an identifiable generating checker, execution command, family-by-family count definition, environment record or source hash for that run.

The search also examined the separately mentioned historical `verify_cm_lm.py` and the earlier audit's 1,536,530-assertion claim. Those claims are not the same run as 20,873, and their named checker was not located either. Later partition-tool strings describe missing artifacts; they are not the missing implementation. Incidental occurrences of 20873 in resource-inventory rows are not evidence about a verification run.

Accordingly, the precise classification is **supported by historical output, not reproducible from the supplied artifacts**. This is not evidence that the old computation was fabricated or that its claims are false. It is a reproducibility defect in the release package. Reconstructing a different loop that happens to count to 20,873 would not repair the provenance.

## New checker and execution

The new `verification/independent_verify.py` does not import the author's computational modules. It uses Python's standard library, exact Boolean operations, explicit tables and independent comparison routes. Its SHA-256 is:

`f2d042ff56f94d5c061f391bd648a5e0e124958792344f9b706a131bae69defc`

Run it from its folder with:

```
python independent_verify.py
```

The recorded run used Python 3.13.5 on Linux and passed **2,022,648 counted equality/characterization checks**. `verification_results.json` records all 28 counter families, their granularities, the environment, source hash and the explicit field `historical_20873_reproduced: false`. This is not a speed benchmark and is not a proof-assistant formalization.

| Domain | What was independently checked |
|---|---|
| All 16 binary connectives | Numeric selection, valuation, normal form, inverse, logical pairing, rotation and algebraic matrix identities. |
| The 16-element formula algebra on two generators | Every reference pair and all external formula pairs, including constants, dependence and contradictory references. The full-formula pairing family alone has 1,048,576 checks. |
| Binary outer operations | All 16 choices each of two inner connectives and an outer connective, with valued LM superposition and pairing preservation. |
| Independent signed input frames | All binary signed-frame combinations in the declared tests, including transport before comparison rather than comparing unaligned arrays. |
| Matched kernel products | All 256 pairs of binary matrices at all eight valuations of the three references. |
| Polarity-image membership | All 65,536 four-cell tensors over the two-generator formula algebra; exactly 16 satisfy the required polarity equivariance. |

| Domain | What was independently checked |
|---|---|
| Partition selectors | All 65,536 four-weight tuples characterized; 256 are partitions. Outer-operation checks then use five specified representative input tensors, not every pair of possible tensors. |
| Ternary functions | All 256 truth functions, valuation/pairing instances and all 48 signed permutations in the declared loops; explicit symbolic transport is separately checked. |
| Flattenings and blocks | Ordered rectangular and noncontiguous selections, and all 16-by-16-by-16 binary block-lift choices across four variables. |
| All 256 two-by-four matrices | Rank-one/conjunctive characterization and minimum XOR-outer-product length, with minimum length computed by breadth-first search independently of Gaussian rank. |
| ANF parity | Every Boolean function through arity four, not just selected truth tables. |
| Compact spectral/algebraic boundary | Cayley–Hamilton and idempotence checks, with eight idempotents and four symmetric idempotents. A compact atlas is supplied as JSON. |

The two-by-four rank distribution is one rank-zero matrix, 45 rank-one matrices and 210 rank-two matrices. The breadth-first decomposition calculation agrees with the rank in every case. The symbolic calculations compare whole formulas represented by truth masks, so they are not merely tests on mutually independent $A, B, X, Y$ variables.

The boundary-witness file records expected failures of overbroad extensions: unaligned fusion, unmatched intermediate frames, unchanged-g output complement, arbitrary nonpartition selectors, OR/XOR interchange, double polarity transport, and a spectral truth-value interpretation of the raw OR matrix. These are not counterexamples to the current qualified theorems.

### What is and is not reproduced

**Reproduced from original generating code:** not the historical 20,873 or 1,536,530 runs. **Independently cross-checked:** the current finite claims and conventions within the explicitly enumerated domains above. **Proved beyond those domains:** the arbitrary-arity identities by the arguments in Section V. **Not established by computation:** historical priority, article significance, physical interpretation, compiler performance or all possible higher-dimensional inputs.

The mandatory manuscript repair is to recover and freeze the old source, or replace the historical assertion with a new, accurately described verification note. The new suite should retain its own count and version rather than being renamed as the old one.

## VII. Independent novelty and prior-art analysis

### Raw numeric operations have close, explicit antecedents

The proceedings version of Cheng, Zhao and Xu's *Matrix Approach to Boolean Calculus* states Proposition 3.3, equation (17), on page 6952: the truth vector of a binary outer operation on two functions of the same ordered variables is obtained entrywise. Reshaping the four entries gives exactly the numeric CM superposition law. This is direct overlap, not merely a broad resemblance to STP. The current manuscript correctly credits it. The surrounding XOR–AND Boolean matrix calculus also prevents presenting the coefficient algebra as new [S1].

Bricken's internally March-1997-dated note explicitly uses the sixteen binary relation matrices and bra–matrix–ket evaluation. Its truth-state convention differs, and some of its calculations use ordinary arithmetic values such as 2 and -1, so it is not safe to identify its entire algebra with the manuscript's XOR–AND ring. It is nevertheless decisive prior art against novelty claims for the binary matrix shape or logical Dirac-style notation. Its verified internal date does not independently establish an original public-release date or peer-reviewed status [S2].

The NPN component is similarly established. Input negation, permutation and output phase are standard transformations of truth functions. Current matrix formulas make those moves visible; they do not create a new equivalence relation or canonicalization problem [S8,S9].

## Formula-valued matrices and the frame mechanism also have close antecedents

Gudder and Latremoliere study matrices over Boolean algebras. Their Example 2.13, specialized to a two-component stochastic vector $(X, \neg X)$, produces the same two-by-two complementary frame pattern used here. Their treatment connects orthogonal Boolean bases and invertible Boolean operators. The scalar addition is join rather than unrestricted XOR, so the transfer must be stated on the orthogonal partition terms where the two agree; arbitrary products in the two calculi are not interchangeable [S3].

This materially narrows the novelty claim. The normal form is a useful application of a classical Boolean frame to a truth-table kernel, rather than compelling evidence of a new kind of matrix. Orthogonal/minterm expansion is also an established subject [S4,S18]. The exact CM/LM labels, reference-frame bookkeeping and combined presentation may be distinctive, but the presence of formulas rather than bits is not enough by itself to establish novelty.

## The central coherence theorem has a simple general explanation

Identify a formula with its truth function on valuations. A formula tensor is then a Boolean function of a valuation and a polarity index. The LM lift is pullback along

$$(x, \alpha) \longmapsto x \Updownarrow \alpha \Updownarrow \mathbf{1}.$$

Valuation fixes $x$; a frame change reindexes inputs; logical pairing restricts the tensor to the index graph selected by the formula states. Each construction preserves Boolean operations. The architecture's coherence is therefore an instance of preservation by function-algebra homomorphisms. This reduction is an independent mathematical comparison, not a claim that a particular source already prints the manuscript's exact diagram. General Boolean-algebra and universal-algebra background supports this perspective [S11,S12].

The strongest possible hostile reading is "several truth-table identities dressed as matrices." The manuscript's answer should not be another repeated assertion that operators are first-class operands. Its stronger answer is an actual demonstration that the type contract catches a frame error, the symbolic representation survives dependent substitution, and one calculation can be carried through multiple semantics without reconstructing its meaning each time. That would establish useful synthesis even if no constituent identity is historically new.

## Spectral logic is a different representation, not a novelty rescue

Eigenlogic uses interpretation-space projectors whose eigenvalues encode truth outputs. The manuscript's diagonal $\mathsf{D}_\Theta$ has this interpretation, whereas the compact two-by-two $\mathbb{F}_2$ CM generally does not [S5,S6]. The appendix correctly distinguishes the two. Earlier Boolean-matrix eigenproblems must also be distinguished from the field spectrum because their scalar operations can differ [S20,S21].

A concrete historical correction is necessary: the inspected passage in Toffano deliberately discussing the avoidance of Dirac notation occurs in **version 12, dated 6 February 2018**, section 5. The first version, posted in December 2015, does not contain that passage. It is sound to cite the work as first posted in 2015, but not to assign the checked version-specific wording to its first version. The separate 2016 Dubois–Toffano paper does use Dirac-style projectors [S5,S6].

## Searches beyond the manuscript's named sources

Orthogonal Boolean expansion under equivalent terminology led to Sule's treatment of Boolean-algebra-valued coefficients and orthonormal expansions [S18]. A 2025 paper on matrix-product-state representations of Boolean functions offers a contemporary comparison involving variable-dependent binary matrices, Shannon expansion and function operations [S19]. These are relevant context for symbolic matrix representations; this audit does not claim that either gives the complete polarity-LM architecture. Matrix-based Datalog literature was also screened but concerns a different relational setting and does not supply a decisive exact antecedent here.

There are limits to the historical coverage. The full Stern monograph was not accessible for comprehensive inspection; the 1992 Mizraji bibliographic detail was not fully independently resolved; Sampei was checked at the primary bibliographic/available indexed-content level, not through a complete page-by-page reading. Exact wording searches did not locate a single earlier source containing the whole present package. **That negative search result does not establish priority.** It supports only the bounded conclusion that the integrated presentation remains potentially differentiated on the inspected evidence.

### Significance is not the same score as novelty

Separate editorial assessments are: new-theorem novelty **4/10**; conceptual usefulness **7/10**; unification value **7/10**; current expository value **6.5/10**; notation/representation value **6.5/10**; methodological potential **7/10**; demonstrated present utility beyond examples **5/10**. These are judgments, not measured quantities, and are distinct from the weighted readiness rubric.

The best genre is a **synthesis/expository article with a methodological core**. A tutorial/reference form is also viable. Calling it a new-results research article would require a stronger case for a substantive theorem or application not already immediate from standard structures. A representation paper need not be rejected merely because it is elementary, but it must earn its notation through clarity, a task or an explanatory advantage.

## VIII. Novelty/priority matrix

The table separates exact overlap, mathematical difference and defensible language. Confidence concerns the indicated comparison, not global firstness. The full discussion and bibliography identify the source-inspection limits.

| Contribution | Closest overlap and remaining difference | Defensible classification and wording |
|---|---|---|
| Two-by-two numeric truth matrix and basis selection | Bricken and broader matrix logic [S2,S13-S16]. State order and naming differ. | **Established prior art; high confidence.** "We use a true-first matrix encoding of binary truth tables." |
| Aligned numeric entrywise superposition | Exact truth-vector law in Cheng et al., Proposition 3.3 [S1]. A reshape does not change the law. | **Established prior art; high.** "The familiar aligned truth-vector law is transported through the LM layer." |
| Formula-valued polarity LM | Boolean-algebra-valued matrices and orthogonal expansions [S3,S4,S18]; the displayed orbit and reference labels organize them in a particular way. | **Known ingredients in a possibly distinctive architecture; medium.** "We organize symbolic polarity orbits and their numeric valuations in one typed calculus." |
| Frame matrix and normal form | The two-component Boolean frame has a close explicit antecedent [S3]; minterm expansion supplies the kernel factorization. | **Straightforward structural formulation; high for mechanism, medium for exact historical formulation.** "The normal form makes reconstruction and specialization explicit." |
| Agreement-bit logical pairing | Boolean state/selector contractions and the frame factorization reduce it to a cofactor identity; numeric bra-ket logic is earlier [S2,S3]. | **Useful representation identity; priority unestablished.** "Pairing has exact agreement-bit semantics." |
| Partition selector characterization | Orthogonal idempotents and a unital linear map yield a Boolean homomorphism [S3,S11]. | **Standard algebraic mechanism; high.** "A selector lemma explains why arbitrary outer Boolean terms commute with pairing." |
| Coherence package | Each map is a Boolean homomorphism or pullback; the combination is the potential contribution. | **Synthesis rather than a demonstrated deep new theorem; high for reduction, medium for architectural distinctness.** "We collect the compatibility laws in a common typed framework." |
| Matched-frame kernel composition | Invertible change-of-basis cancellation; parity kernel composition is ordinary matrix multiplication. | **Straightforward corollary; high.** "The representation also transports parity kernel composition when the middle frame matches." |

| Contribution | Closest overlap and remaining difference | Defensible classification and wording |
|---|---|---|
| Orbit image and equivariance | An orbit-valued equivariant object is determined by its reference cell [S11]. | **Standard result in concrete notation; high.** "Equivariance characterizes the LM image." |
| Tensors and rectangular flattenings | Truth tables, one-hot tensors and coordinate reshaping [S7,S10,S19]. The typed operand labels are the specific emphasis. | **Standard representation, potentially useful specification; high.** "The construction extends to ordered rectangular correspondence maps." |
| XOR/conjunctive rank decomposition | Ordinary field rank equals minimum outer-product length. The Boolean-function interpretation changes no rank theorem. | **Standard linear algebra; high.** "Flattening rank quantifies XOR-conjunctive separability across the declared partition." |
| Signed frames/NPN | Established truth-function transformations and implementation reuse [S8,S9]. | **Established prior art; high.** "We express standard input-frame transformations in the CM/LM notation." |
| ANF top coefficient and truth parity | Direct Boolean Mobius inversion [S10]. | **Standard corollary; high.** "Truth-tensor parity detects the full-degree coefficient." |
| Compact spectra and diagonal truth observable | Elementary finite-field classification versus established spectral truth operators [S5,S6,S20,S21]. | **Boundary/expository contribution; high.** "Compact CM spectra must be distinguished from interpretation-space truth spectra." |
| Entire integrated paper | No inspected source was shown to contain the identical package; inaccessible literature remains. | **Apparently differentiated organization, not certified priority; medium.** "A unified formula-valued symbolic/numeric presentation with explicit operation and frame contracts." |

# IX. Reviewer-feedback closure matrix

The older foundation audit addresses an earlier revision. Its criticisms were checked against the **current** source rather than copied into this verdict. The compiler reviews were also separated from foundations-specific obligations. A later register repeating a criticism is not an additional independent audit.

| Feedback | Disposition and current location | Independent assessment |
|---|---|---|
| F01: Earlier CM_LM_Audit_Astra: zero/one-triggered mask mismatch | **Resolved**. Theorem 6.4; equation (36) | The current pi mask is explicitly one-triggered and the valuation formula agrees. Do not repeat the old counterexample as a current defect. |
| F02: Earlier audit: free references versus substituted formulas | **Resolved**. Section 3.1; Theorem 3.2 | Joint satisfiability is now confined to all-true specialization. Frame inversion remains valid even without it. |
| F03: Earlier audit: introduce invertible formula frame normal form | **Resolved**. Theorem 3.1 | Both forward and inverse identities and their minterm proof are present. |
| F04: Earlier audit: pairing preservation needs a structural proof | **Resolved**. Theorem 4.5; Theorem 6.12 | Orthogonal partition weights now explain arbitrary outer Boolean operations. |
| F05: Earlier audit: term extension for arbitrary f | **Resolved**. Equation (31); Definition 6.5 | The canonical Boolean term is explicit, so formula substitution is well typed. |

| Feedback | Disposition and current location | Independent assessment |
|---|---|---|
| F06: Earlier audit: exact higher-arity pairing and reconstruction | **Substantially resolved**. Theorem 6.15 | The theorem is correct. Expand the sentence identifying the selected tuple as agreement bits. |
| F07: Earlier audit: flattened noncontiguous variable order | **Resolved**. Section 7; Definition 7.1 | An ordered assembly map removes the ambiguity. |
| F08: Earlier audit: rank-one converse and minimum XOR rank | **Resolved**. Theorem 8.2 | Both directions and the rank-factorization argument are present; only the zero/empty-sum convention needs explicit wording. |
| F09: Earlier audit: top-degree statement must exclude zero arity | **Resolved**. Proposition 10.1 | The current proposition explicitly assumes n>=1. |
| F10: Earlier audit: input/output phase distinction | **Resolved**. Theorem 6.12(3); Remark 6.13 | Output complement is excluded from the unchanged-g law and the dual function is supplied. |
| F11: Earlier audit: distinguish ordinary LM multiplication | **Resolved**. Section 5.1; Theorem 5.5 | Matched intermediate frames are required; the text distinguishes parity kernel composition from other products. |
| F12: Earlier audit: action direction | **Substantially resolved**. Section 9 | The numeric right action is exact. An explicit symbolic signed action should still be added. |
| F13: Earlier audit: elementary orbit result is not a novelty theorem | **Resolved as wording**. Remarks 6.8, 6.10 and Section 13 | The manuscript now acknowledges the classical function-algebra mechanism. The value of the remaining synthesis still needs demonstration. |
| F14: Earlier review: pointwise numeric superposition already in Cheng | **Resolved as attribution**. Introduction; Section 5; Table 1 | The 2011 Proposition 3.3 is explicitly credited and independently confirmed in the proceedings PDF. |
| F15: Earlier review: Bricken predates bra-ket and 2 by 2 matrices | **Substantially resolved**. Introduction; Section 13 | The direct numeric antecedent is acknowledged. Retain the distinction between internal date and verified public availability. |
| F16: Earlier audit: stronger Boolean-space/orthogonal-expansion comparison | **Partially resolved**. Section 3.2 | Gudder and Sampei are cited, but the exact n=2 frame construction in Gudder–Latremoliere deserves a direct comparison. |
| F17: Earlier audit: do not confuse raw spectrum and truth values | **Resolved**. Appendix A | The OR/XOR examples, diagonal observable bridge and absence of physical measurement semantics are now clear. |
| F18: Earlier audit: phase and quantum material overload | **Substantially resolved**. Section 11; discussion; Appendix A | The extra multiplication is fenced off. The spectrum appendix could be shortened further for exposition. |
| F19: Earlier audit: diagrams need layout work | **Substantially resolved**. Figures 1-4, compiled pages 6,16-18 | No clipping found; Figure 2 upper label is still crowded against its boxes. |
| F20: Earlier audit and revision notes: many repeated OR/AND/XOR demonstrations | **Partially resolved**. Examples 5.3-5.4; Remark 6.14 | Current examples are correct but should be threaded into one calculation instead of reset repeatedly. |
| F21: Preservation audit, viability review and revision notes: runnable foundations checker | **Unresolved in supplied package**. Appendix E | The missing generating source is independently confirmed. New verification supplied by this audit is a replacement, not recovered provenance. |
| F22: Earlier audit: 1,536,530 checks and verify_cm_lm.py | **Requires external evidence**. Historical audit finite-verification section | A historical assertion is present; its named program is not located. This neither validates nor disproves that past run. |

| Feedback | Disposition and current location | Independent assessment |
|---|---|---|
| F23: Earlier literature discussion: Eigenlogic deliberately avoids Dirac notation | **Partially resolved; version correction needed**. Introduction; Toffano bibliography entry | The inspected supporting passage is in v12 (2018), not the first 2015 version. Credit the work and version separately. |
| F24: Bibliographic placeholder for Zhao/Gao/Cheng | **Unresolved**. Bibliography ZhaoSTP | The undated author preprint is identifiable. Supply that stable record, or compare it with the related 2012 journal paper before replacing the citation. |
| F25: Viability review: justify standalone value/genre with a task | **Unresolved**. Introduction and discussion | Typed architecture is plausible but an end-to-end demonstration and explicit comparison remain the highest-value journal revisions. |
| F26: Compiler R1/R2: hybrid failure, benchmark freeze, speed and comparator obligations | **Not applicable to foundations verdict**. Actual compiler manuscript and reviews | Those are compiler-specific obligations. The foundations manuscript uses none of those empirical claims in its proofs. |
| F27: Compiler R1: scalar transpose/axis confusion in earlier computational presentation | **Obsolete for this source**. Equation (6); Example 5.3 | The current foundations transpose and ordered-frame statements are correct. |
| F28: Portfolio warnings: missing figures in other papers | **Not applicable to this source**. Fresh foundations build | This source defines its figures inline with TikZ and compiles without missing assets. Do not import another manuscript build failure. |
| F29: Repeated later registers endorse prior findings | **Not independent verification**. Program partition tools and summary registers | Their repetition is provenance/triage evidence, not additional mathematical or experimental confirmation. No evidence of intentional misrepresentation is inferred. |

The substantive unresolved foundation issues are therefore not the earlier mask error or a failure of dependent-formula semantics. They are the runnable-checker claim, exact historical versioning and citations, the residual symbolic transport specification, and the need to demonstrate the value of the synthesis. No evidence was found that warrants an allegation that earlier reviewers intentionally misrepresented their work. Their stated inspection limits and the provenance of repeated conclusions should simply remain visible.

# X. Bibliography and historical-attribution audit

There are **21 bibliography entries** in the audited source. The following status table addresses all of them. "Metadata checked" does not mean the complete work was read. The source register at the end of this report identifies the accessible primary records and important locators.

| Manuscript entry | Audit status | Required or recommended action |
|---|---|---|
| Edwards 1972 | Publisher metadata supports title, journal 15(3), 247-253 and year. Full paper not comprehensively inspected [S13]. | Retain; DOI 10.1093/comjnl/15.3.247 may be added. Do not use a broad citation as evidence for a specific LM theorem. |
| Stern 1988 | Book/preview record supports author, matrix-logic work and publication. Full monograph not reviewed [S14]. | Retain conservatively; do not claim that an unsearched book lacks formula-valued precedents. Verify the edition's full title on its title page before submission. |
| Mizraji 1992 | Exact primary bibliographic/content verification remains incomplete in this audit [S15]. | Independently confirm title, volume/issue/pages against the original record. Do not add an unverified DOI. |

| Manuscript entry | Audit status | Required or recommended action |
|---|---|---|
| Mizraji 2008 | Publisher issue record supports 18(1), 97-121 and DOI 10.1093/logcom/exm057; an author manuscript was available [S16]. | Retain 2008 as issue year; distinguish an earlier online date from that year. |
| Cheng and Qi 2010 | Author PDF/record supports the IEEE article, 55(10), 2251-2258 [S7]. | Retain. This is numeric structure-matrix context, not proof that the full LM architecture already appears there. |
| Cheng, Zhao and Xu 2011 | Actual proceedings PDF and Proposition 3.3 checked [S1]. | Retain exact proceedings version, pages and DOI. The manuscript's direct attribution is appropriate. |
| ZhaoSTP | The named nine-page author-hosted preprint exists. The "details to be checked" wording is not publication-grade [S22]. | Cite it accurately as an undated preprint with stable URL/access date, or first compare it with the related 2012 journal article before substituting that record. |
| Toffano 2015 | First-posting date is correct, but the inspected Dirac-notation statement belongs to v12 (2018), not v1 [S5]. | Cite the first-posted work and the specific 2018 version/section supporting that assertion. |
| Dubois and Toffano 2016 | arXiv:1607.03509, authors, year and projector/Dirac content checked [S6]. | Retain as the preprint actually inspected; do not invent journal publication metadata. |
| Droncheff 2018 | Public record distinguishes May 2018 labeling and upload on 1 October 2018 [S17]. | The current caution is correct. Retain the historical author name and DOI; do not collapse the two dates. |
| TheoryCompiler2026 | Unpublished companion exists in the archive and actual relevant manuscript sections were inspected. | Give a stable deposited version or a precise unpublished description. Keep the foundations proofs self-contained. |
| TheoryPhase2026 | In-preparation self-citation is not an independently retrievable finalized publication. | Remove if unused; otherwise retain explicitly as in preparation, without making it a necessary proof reference. |
| Rutherford 1965 | Publisher metadata supports 67(1), 25-38 and DOI 10.1017/S0080454100010712 [S20]. | Retain the warning that its Boolean algebra need not be the present XOR–AND field algebra. |
| Blyth 1967 | Publisher metadata supports 67(3), 196-204 and DOI 10.1017/S0080454100008049 [S21]. | Same algebraic qualification; no full-paper theorem comparison is claimed here. |
| Bricken 1997 | Actual nine-page note inspected; internal date and explicit binary bra-ket examples confirmed [S2]. | Retain the unresolved public-release/peer-review qualification. Distinguish its mixed arithmetic examples from the present scalar convention. |
| Mishchenko et al. 2006 | Proceedings PDF/record supports three authors, DAC pages 532-536 and DOI 10.1145/1146909.1147048 [S8]. | Retain as NPN/cut-reuse context, not a direct formula-LM source. |
| Zhou, Wang and Mishchenko 2019 | Author PDF confirms the three-author TRETS article 12(2), Article 7, 16 pages; DOI 10.1145/3313917 [S9]. | Current authors are correct. Do not confuse it with a related four-author conference paper. |
| Gudder and Latremoliere 2009 | Primary paper and arXiv publication record checked [S3]. It also appeared in Linear Algebra and its Applications 431(1-2), 272-296. | Strengthen to the journal citation while retaining the accessible preprint. Add the exact frame/orthogonal-basis comparison. |
| Sampei 1950 | Primary metadata supports author, pages 113-125, volume 11(3) and DOI 10.14492/hokmj/1530864050 [S4]. Full article access was limited. | Retain as historical orthogonal-expansion context; avoid a more specific theorem claim without reading the relevant page. |

| Manuscript entry | Audit status | Required or recommended action |
|---|---|---|
| Burris and Sankappanavar 1981/2012 | Author-hosted corrected 2012 edition identifies the original 1981 book [S11]. | Current edition distinction is sound. Give a relevant chapter/section if making a precise attribution. |
| Post 1941 | Publisher preview identifies the work and original 1941 copyright/series [S12]. | Retain 1941; a later reprint/catalog year is not grounds to silently replace it. |

The clearest required changes are the Zhao placeholder and the version-specific Toffano attribution. Strengthening the Gudder comparison matters more than adding a long list of distant operator papers. Verifying the outstanding Mizraji record is a small but necessary submission task. The 2025 MPS comparison is useful optional context, not a mandatory claim that the manuscript has been superseded.

No unpublished companion is needed to follow the core proofs as written. A scholarly citation may acknowledge origin or a related program, but it should not be the only source of a hypothesis, definition or proof used in this paper. The current foundations manuscript largely satisfies that requirement.

# XI. Scope and cross-paper boundary analysis

## Foundations versus compiler

The actual companion compiler manuscript contains the binary formula LM, its valuation, logical pairing and valuation/pairing commutation in its LM section. These are genuine overlapping mathematical passages, not overlap inferred solely from an ownership register. Its implementation discussion also makes clear that the compiler operates on numeric four-bit CM tokens and pair metadata, not on arbitrary symbolic formula-valued matrices. Therefore the foundations calculus should not be credited with an implemented formula-level compiler that the companion does not actually implement.

This overlap is manageable. The foundations article should own the full definitions and proofs. The compiler can retain a short, self-contained statement of the numeric frame/fusion invariant, with a concise explanation of its relation to the LM viewpoint and a precise cross-reference. It should not be necessary to read a performance paper to understand Theorem 4.1 here, nor to reproduce every foundational theorem in the compiler.

The companion's unfinished confirmatory experiment and the old hybrid/fallback review issues are **not mathematical defects in this foundations manuscript**. No speed or memory claim is used as a premise here. Conversely, the correctness of this calculus is not evidence of compiler profitability. The report does not certify the companion's source, experimental protocol or performance.

## Foundations versus phase and modal work

The archive's correctness/model-boundary report explicitly separates native two-by-two CM logic, a rotation-generated coefficient algebra, shared modules, literal tensor systems and additional allowed gates/measurements. Selected actual source passages were inspected, not just the ownership summaries. Those distinctions reinforce the current foundations boundary: truth-table selection alone does not establish Bell phenomena, teleportation, interference resources or a physical theory.

The current manuscript correctly identifies literal rotation as a signed coordinate action and treats cyclic convolution as an additional multiplication. Its short group-algebra isomorphism is mathematically sound, but the development of units, nilpotents, operational restrictions and modal resources belongs elsewhere. Nothing found in those companion corrections invalidates the formula frame or agreement pairing. No modal correctness campaign was rerun as part of this foundations audit.

The spectral appendix can remain as a defensive clarification, but it should not grow into another paper inside this one. A compact explanation that matrix elements and eigenvalues are different semantics, with OR as a counterexample and the diagonal truth representation as a comparison, is enough for most readers.

## Structural computation and decomposition

The structural-computation collection describes research endpoints and incomplete empirical work, not a completed experiment supporting the present foundations. The ordinary rank/separability theorem belongs

naturally here as a mathematical explanation of flattenings. Claims about avoiding materialization, structural retrieval, persistent decomposition or end-to-end speed belong to the structural/computational work and require their own evidence.

**Standalone assessment:** the current foundations mathematics is substantially self-contained. The remaining cross-paper task is editorial ownership and reduction of duplicated proofs, not recovery of an external theorem without which this paper fails.

# XII. Completeness audit

The main mathematical objects, scalar algebra, valuation map, polarity convention, ordered flattening and pointwise operation are defined. The proofs needed for the principal claims are present. The theorem count does not conceal a missing central result. The following changes would make the paper more complete as a standalone article without enlarging its research scope.

First, make the symbolic signed-frame action explicit using the identity in Section V of this report. This closes the gap between an informal "typed" promise and an executable convention. Second, explain the higher-pairing selector step rather than relying on an abrupt appeal to a general theorem. Third, state the zero/empty-sum and optional empty-block conventions where relevant.

Fourth, supply one complete worked example from a logical expression through two differently framed occurrences, alignment, operator combination, symbolic pairing and numeric valuation. The current short examples each illustrate a correct local identity, but the reader does not yet see the whole promised architecture doing one job. A small noncontiguous rectangular example would also help readers use the assembly definition.

Fifth, publish the verification source actually supporting the new revision and accurately state what was checked. The present verification appendix is incomplete as an artifact claim, despite the finite mathematics being independently sound.

Finally, align headings with content. Appendix A is titled a compact-CM atlas, but presents representative spectral examples and the idempotent classification rather than a full sixteen-row atlas. Either rename it to "Spectral boundary and examples" or supply a genuine complete table as supplementary material. The independent atlas JSON in this audit can support the latter after integration and review.

The binary LM basis appendix is useful but could be shortened by referring to the normal form. The compatibility appendix largely repeats the main theorem and could become a one-page reference box. The notation table is valuable as an appendix. The finite verification note should stay separate from the proofs, with its executable supplement clearly linked. No full clone formalism, operator-valued enrichment, new phase theory, or physical measurement axioms are needed to make the present article complete.

# XIII. Exposition and editorial review

## Organize around the useful mechanism

The reader should encounter a practical semantic problem before a long list of representation levels. A good opening is the reversed-implication example: two local operators have identical numeric arrays, but canceling them without their operand frames gives the wrong answer. This motivates labelled frames. The formula frame then explains how to transport them; the selector theorem explains why pairing and outer operations agree.

The strongest short narrative is: **truth kernel; symbolic frame; agreement pairing; Boolean-homomorphism coherence; typed higher-arity extension**. Ordinary rank and rotation boundaries are applications/reference material, not competing main themes.

The introduction, abstract, status paragraph, related-work section, discussion and conclusion repeatedly explain that the individual ingredients are classical and that no physics or speedup is assumed. Those cautions are appropriate, but repetition makes the article sound defensive and interrupts the mathematics. Give one precise contribution statement, one operation/type table and one boundary paragraph; use short cross-references thereafter. Removing repeated disclaimers is not permission to strengthen the novelty claims.

## Concrete section-level edits

| Location | Editorial action | Reader benefit |
|---|---|---|
| Abstract | Lead with the frame normal form and agreement semantics; shorten the inventory of extensions. | The organizing idea becomes visible before the notation list. |
| Introduction/status paragraph | Replace project-history prose with a short provenance note and one clear contribution boundary. | A reader outside the portfolio can understand the paper without knowing the earlier split. |
| Section 2 | Retain the three-composition distinction; consider making several elementary propositions a compact preliminary lemma. | Fewer theorem labels compete with the actual structural explanation. |
| Section 3 | Show one LM explicitly factored by its two frames, including a dependent-reference specialization. | Explains what is retained beyond a single reference formula. |
| Section 4 | State in words that pairing asks the connective about agreement with two references, not about the original truth values directly. | Avoids a physical or probabilistic interpretation. |
| Sections 5-6 | Reuse the same example through superposition, valuation and pairing. | Demonstrates coherence rather than repeating isolated identities. |
| Section 6.5 | Add the valuation/selected-index calculation from this report. | Removes the most compressed proof transition. |
| Sections 7-8 | Add one two-by-four noncontiguous flattening example; state the limitation of one-bit block summaries. | Prevents a reader from mistaking a conditional block lift for universal compact decomposition. |
| Section 9 | Add the explicit symbolic signed-frame formula and one mask/permutation check. | Makes the type contract auditable and implementable. |
| Section 10 | Fix the row vector $t$; shorten the cyclic-algebra teaser. | Prevents a dimension mismatch and keeps the coordinate comparison focused. |
| Section 11 | Keep the distinction between the four-coefficient rotation operator and a four-variable map; move most future phase discussion out. | Equal matrix size no longer suggests equal semantic type. |
| Related work | Compare the exact Boolean-frame construction, not just fields of study. | Gives a fair account of what is and is not new. |
| Discussion/open directions | Shorten the six-item expansion agenda or move it to supplementary research notes. | Avoids making the article look like a prospectus for several unfinished papers. |
| Conclusion | State what the reader can now calculate correctly and what remains outside scope. | Ends with an outcome rather than another comprehensive disclaimer. |

The nonstandard $\Updownarrow$ notation is internally explained and need not be replaced. “Superposition” is defensible in the defined Boolean-function sense but should normally appear as “pointwise Boolean superposition.” “State,” “bra” and “ket” are acceptable shorthand once the formula-state and Boolean-algebra meaning is fixed. “Logical matrix” is overloaded in existing literature, so retain the qualifier **formula-valued LM** at first use in each main context. “Spectrum” needs its field and multiplicity convention when developed further.

A shorter article does not require deleting the conceptual contribution. It requires giving the reader fewer parallel names for the same preservation mechanism and more evidence of why that mechanism is useful.

# XIV. Typesetting and build audit

The unchanged source was compiled from scratch using pdfLaTeX, with shell escape disabled, and a second pass resolved cross-references. The source is self-contained with its bibliography and TikZ definitions; no external image assets were required.

```
pdflatex -interaction=nonstopmode -halt-on-error -no-shell-escape \
  -output-directory=build source/audited_foundations.tex
# Run the same command a second time.
```

The fresh output is **31 pages, 615,830 bytes** in the recorded build. The pass logs contain no unresolved citations, undefined references, duplicate-label diagnostics, or overfull/underfull-box warnings. The shell-escape-related epstopdf warning is benign for this source, which needs no EPS conversion. The supplied PDF has the same page count and the same extracted text on every page. PDF byte identity is not claimed.

All pages were rendered and visually surveyed. Figures 1-4 were checked in the rendered document, including larger views of pages 6, 16 and 18. No clipped text, unreadable mathematical glyphs, missing figures or margin overflow was found. Page 16's Figure 2 upper arrow label consumes almost the full gap between boxes; splitting "LM operator superposition" into two lines would avoid its white label background crowding the box borders. This is an editorial defect rather than a failed build.

The related-work table on page 24 and notation table on page 29 are dense; they are legible but could benefit from a shorter text column or slightly more space. Some proofs begin on the page after their theorem statement. The last bibliography page is sparse. These are P2/P3 presentation matters, not evidence that the current equations are broken.

The audit did not certify PDF accessibility tagging, every external hyperlink destination or identical rendering in every PDF reader. Source/PDF consistency, successful compilation and the visual checks above are the demonstrated build results. The logs and freshly compiled paper are included in the bundle.

# XV. Publication blockers

**P0: none established in this audit.** No central false theorem, missing core argument, unavoidable contradiction of types or unsupported principal firstness assertion was found. The verification run is ancillary, so its missing original source should not be escalated into an alleged failure of the central mathematics. It still must be corrected as a factual publication claim.

| ID / level | Location and problem | Required repair and effect |
|---|---|---|
| B1 / **P1** | Appendix E, L1473-1474: a checker is said to accompany the revision, but the generating artifact for 20,873 is not located. | Recover it with a count definition and manifest, or replace/withdraw that historical reproducibility assertion and ship the new checker under its own identity. Required before release. No change to the theorems. |
| B2 / **P1** | Introduction and bibliography: unresolved Zhao entry, version-specific Eigenlogic attribution, incomplete verification of one older bibliographic record. | Remove the placeholder; accurately cite the inspected Toffano version; check the outstanding Mizraji metadata. Required before submission and the concrete known corrections before preprint. No mathematical conclusion changes. |
| B3 / **P1** | Contribution statement and related work: the closest frame/Boolean-space mechanism is not compared directly enough. | Explain the exact overlap and the restricted OR/XOR transfer; position the contribution as synthesis/methodology unless a stronger result is established. Important for journal review, not a demand to invent new mathematics. |
| B4 / **P1 for a specification article** | Theorem 6.12(3) and Section 9: symbolic combined transport is less explicit than the advertised typed architecture. | Add the identity in Section V and the double-mask warning. A small edit; no central theorem needs to be withdrawn. |
| B5 / **P1 for journal submission** | Introduction, examples and discussion: usefulness of the integrated representation is asserted more than demonstrated. | Add one complete task/example and consolidate the repeated narrative. Establishes the article's value for its chosen genre. |
| B6 / **P2** | Section 10, L1167-1175: row truth vector and column vec conventions conflict. | Set $t = \mathrm{vec}([\Theta])^T$. Fix before preprint; local dimensional repair only. |
| B7 / **P2** | Theorem 6.15 proof; Theorem 8.2; optional empty blocks. | Expand the selected-index step and state empty-product/XOR conventions. Clarifies generality and degeneracies. |

| ID / level | Location and problem | Required repair and effect |
|---|---|---|
| B8 / **P2** | Foundations/compiler duplication and project-history prose. | Give full proof ownership to the foundations and shorten companion repetition. No need to merge the papers. |
| B9 / **P2** | Appendix A title promises an atlas more complete than the displayed examples. | Rename it or include a full supplementary atlas. The existing spectral claims remain correct. |
| B10 / **P3** | Figure 2 label, dense reference tables, theorem/proof page splits, sparse final page. | Small TikZ and pagination edits. Cosmetic/readability effects only. |
| B11 / **P3** | Formula-matrix convention says "this subsection only"; repeated compatibility prose. | Clarify that the product convention also governs the matched-frame theorem and shorten repetition. |

A preprint can be made accurate through a short correction pass even though the journal-level exposition and contribution case require major revision. Those are different thresholds; the weighted score does not erase either one.

# XVI. Smallest revision program to reach the next readiness state

## Mandatory correctness and factual changes

Freeze a separately attached source file and record its hash. Correct the STP vector orientation. Add the explicit symbolic transport identity and a sentence making clear that a mask is applied once. Expand the higher-pairing proof by identifying the selected polarity tuple. State the zero-sum convention in the rank theorem. None of these requires changing the normal form, pairing theorem or arbitrary-arity architecture.

## Novelty and claim calibration

Replace repeated claims that the integration is distinctive with one precise paragraph explaining the representation and its compatibility. Add the exact Gudder–Latremoliere comparison and retain the direct Cheng attribution. Describe the article as a formula-valued representation calculus with expository/methodological value. Do not claim that all formula-valued Boolean matrices, bra-ket logic, arbitrary pointwise operator operations or invertible Boolean frames originated here.

## Literature and history

Resolve the Zhao placeholder using the source actually cited. Correct the Toffano version attribution. Strengthen the Gudder citation to its journal publication and verify the remaining old bibliographic metadata. Preserve the manuscript's careful distinction between Droncheff's May label and October upload and between Bricken's internal date and public chronology. No exhaustive history of every matrix-logic formalism is needed, but close mechanisms require accurate attribution.

## Reproducibility

Choose explicitly between recovery and replacement. With replacement, publish the new checker, outputs, environment, source hash, exact enumeration domains and counted units. Update Appendix E to say which finite families were tested and make clear that the proofs handle arbitrary arity. Remove the assertion that the old 20,873 run is reproducible unless its original evidence is actually recovered. Do not retain two unexplained large counts as interchangeable endorsements.

## Structure and exposition for journal submission

Use the reversed-implication calculation as the entry point, then continue one expression through normal form, pairing and valuation. Include dependent references to show what the full LM preserves. Shorten the introduction's project history, merge redundant elementary results, and consolidate the claim boundary. Keep phase multiplication and modal resources outside the main development. Assign full foundational proof ownership here and shorten the compiler's duplicate section.

### Optional, not prerequisites

A full sixteen-CM reference atlas, a small noncontiguous flattening example, a compact formula-pullback diagram, or a short comparison with contemporary Boolean matrix-product representations would help. A full clone formalization, new universal categorical language, operator-valued enrichment, additional phase theorems and an expanded quantum discussion are not needed for the next release and risk recreating the scope problem.

**Acceptance test for the revised preprint:** source/PDF identity is frozen; all known factual/type corrections are made; the verification note matches a shipped runnable artifact; bibliography placeholders are gone; the claim paragraph states the honest genre. **Acceptance test for a journal-ready revision:** additionally, the paper demonstrates why its architecture is useful and reduces duplicated or peripheral material enough that the contribution is easy to identify.

## XVII. Final readiness determinations and explicit answers

### Five distinct determinations

**A. Public preprint: READY AFTER MINOR CORRECTIONS.** The mathematics is sufficiently sound for public circulation once the known artifact, citation-version and local typing issues are corrected. This is not approval to post the unmodified verification claim. A preprint can accurately present a synthesis without first proving a new research theorem.

**B. Journal submission in current form: MAJOR REVISION REQUIRED.** The remaining work concerns the contribution case, comparison with close antecedents, demonstration of utility, reproducible supplement and narrative focus. A journal-facing reader should not have to infer the genre from repeated caveats.

**C. New-results research article: NOT READY on the evidence established here.** No substantial new theorem has been independently distinguished from the standard Boolean-frame, function-algebra and rank mechanisms. This does not rule out publication in a venue that values representation or synthesis, nor does it prove that no novel theorem could be developed later.

**D. Expository/synthesis article: the strongest defensible publication route; revision required.** The representation can clarify several operations and prevent real alignment mistakes. It needs a more economical narrative and a complete example to realize that value. Its classical ingredients are not, by themselves, a reason against publication.

**E. Technical/methodological article: plausible but not ready in its current presentation.** A fully explicit frame transport contract, operation/type signatures, conformance tests and a demonstrated task would support this genre. The current code independently checks the identities, but the paper should not imply an implemented symbolic compiler or performance benefit.

### Answers to the 25 requested questions

| No. | Question | Answer |
|---|---|---|
| 1 | Are central statements false? | None found in this audit. Expected failures of stronger extensions are identified separately. |
| 2 | Are central statements true only after adding hypotheses? | The principal needed hypotheses are already present. Symbolic transport needs a more explicit definition, not rescue of a false central theorem. |
| 3 | Are arbitrary-arity results genuinely proved? | Yes. Pullback, Boolean term, partition and rank arguments apply for every finite arity under their scopes. |
| 4 | Is the formula-valued construction well typed? | Yes, in the declared formula algebra. Free-reference equivariance should not be silently exported to a constrained quotient. |
| 5 | Is pairing correctly characterized? | Yes: it applies the connective to reference-agreement bits. It is not a physical measurement or probability. |
| 6 | Does valuation commute as claimed? | Yes for the stated Boolean term, contraction and pointwise operations. All-true specialization has its explicit satisfiability condition. |

| No. | Question | Answer |
|---|---|---|
| 7 | Are pointwise and matrix products separated? | Yes. The matched-frame product is separately defined as parity kernel composition. |
| 8 | Is rank over the correct algebra? | Yes, ordinary $\mathbb{F}_2$ rank, not OR-based Boolean rank. Clarify the zero/empty-XOR convention. |
| 9 | Are signed frames and NPN accurate? | The numeric action and NPN moves are correct. Add the exact combined symbolic transport formula. |
| 10 | Are spectra appropriately limited? | Yes. Compact field spectra, Boolean-algebra spectra and diagonal truth projectors are distinguished. |
| 11 | Is the manuscript standalone? | Substantially yes mathematically; the verification artifact and bibliography need completion. |
| 12 | Is 20,873 reproducible? | No, not from the supplied generating artifacts. The preserved log is not a runnable checker. New independent checks are not that historical run. |
| 13 | Which previous criticisms remain? | Checker recovery, exact attribution/metadata, stronger mechanism comparison, explicit transport and demonstration of usefulness. |
| 14 | Which are obsolete or incorrect now? | The old polarity-mask mismatch, an unqualified all-true valuation, missing canonical extension/assembly, and missing rank converse are repaired. Compiler-only requirements do not apply here. |
| 15 | Which contributions are prior art? | Numeric matrix logic, bra-ket evaluation, aligned truth-vector composition, Boolean orthogonal frames, NPN, Mobius parity and ordinary rank factorization. |
| 16 | What appears differentiated? | The particular integrated formula-polarity representation with its explicit valuation, agreement pairing and operation/frame organization. Priority is not established. |
| 17 | Is new-theorem novelty sufficient? | Not demonstrated under a significant-new-mathematics research-article expectation. |
| 18 | Is another publishable form plausible? | Yes: focused synthesis/exposition or a rigorously illustrated methodological specification. No acceptance prediction is made. |
| 19 | Are title and abstract calibrated? | The title is defensible but broad. The abstract is largely cautious; it should foreground the useful mechanism and reduce the inventory of claims. |
| 20 | What is understated? | Reconstruction of the connective from the entire LM survives dependent/contradictory reference substitution even when the reference cell alone collapses. |
| 21 | What needs narrowing? | Any implied novelty of formula-valued matrices or first-class operator treatment in general, and any suggestion that a commuting diagram alone supplies a deep new theorem. |
| 22 | What belongs elsewhere? | Full phase multiplication, modal resources, performance/benchmark claims and extended structural-computation research. |
| 23 | Exact P0/P1 blockers? | No established P0. P1: missing checker claim, bibliographic/version corrections, close-mechanism/genre positioning, specification-level transport and journal-level utility demonstration. See B1-B5. |
| 24 | Ready for preprint? | After the identified minor correction pass, not with the uncorrected verification and citation wording. |
| 25 | Ready for journal submission? | No; major revision of the presentation, contribution case and supporting release package is required. |

# XVIII. Suggested contribution language

## Replacement contribution paragraph

We present a formula-valued representation calculus for Boolean functions with explicit reference frames. A logical matrix is the polarity orbit of a reference formula, and its frame normal form relates it to the corresponding numeric truth matrix by invertible Boolean matrices. Logical pairing evaluates the represented connective on agreement with the reference formulas. These constructions fit into a common system of

compatibility laws: valuation, aligned pointwise Boolean composition, input-frame transport and pairing preserve the same Boolean operations. The calculus extends to arbitrary-arity truth tensors and ordered rectangular flattenings, where ordinary field rank measures XOR-conjunctive separability across a chosen variable partition. The aim is a precise and usable symbolic/numeric organization of classical Boolean mechanisms, rather than a new Boolean algebra, physical measurement theory or compression scheme.

### Replacement novelty/related-work paragraph

Truth-vector composition, logical matrix evaluation, Boolean orthogonal expansions, input-frame transformations and matrix rank factorization have substantial prior literature. In particular, aligned numeric pointwise composition is already explicit in Cheng, Zhao and Xu, while Boolean-algebra-valued frame constructions provide a close antecedent for the complementary frame matrices used here. The contribution pursued in this paper is the integrated reference-polarity presentation and its explicit agreement-pairing and transport contracts. We therefore distinguish the utility of this formulation from claims of historical priority for its ingredients; no firstness claim is made for formula-valued matrices, logical bra–matrix–ket notation or pointwise operations on logical operators in general.

These paragraphs are suggested replacement language, not a rewritten manuscript or a claim that the requested revision has already been applied. The delivered source remains unchanged.

## Primary-source register and audit evidence

The following sources were used at the indicated depth. URLs are supplied for reproducibility. A primary record or preview may verify metadata without supporting a full mathematical comparison. Retrieval and comparison date: 26 September 2026. Sources unavailable in full remain limitations rather than silently completed readings.

**[S1] Cheng, Zhao and Xu (2011).** *Matrix Approach to Boolean Calculus*, CDC-ECC, pp. 6950-6955, DOI 10.1109/CDC.2011.6160289. Actual proceedings PDF inspected; key locator Proposition 3.3, equation (17), p. 6952. Source: https://skoge.folk.ntnu.no/prost/proceedings/cdc-ecc-2011/data/papers/0224.pdf.

**[S2] Bricken, William.** *Notes on Matrix Techniques for Logic*, internally dated March 1997. Nine-page technical note inspected, including binary matrices, bra-ket expressions and arithmetic examples. Public-release chronology is not established by that internal date. Source: https://wbricken.com/pdfs/01bm/01math/03math-supporting/math-tangential/04matrix-tech.pdf.

**[S3] Gudder, Stan, and Frederic Latremoliere (2009).** *Boolean inner-product spaces and Boolean matrices*, Linear Algebra and its Applications 431(1-2), 272-296; arXiv:0902.1290. Primary paper and publication record inspected. Important locators: Example 2.13, Lemma 2.15 and Theorem 4.1. Source: https://arxiv.org/abs/0902.1290; paper: https://arxiv.org/pdf/0902.1290.

**[S4] Sampei, Yoemon (1950).** *On the orthogonal expansion of the Boolean polynomial and its applications I*, Journal of the Faculty of Science, Hokkaido University, Series I, 11(3), 113-125; DOI 10.14492/hokmj/1530864050. Primary metadata and available indexed content; full-page access limited. Source: https://doi.org/10.14492/hokmj/1530864050.

**[S5] Toffano, Zeno.** *Eigenlogic in the spirit of George Boole*, arXiv:1512.06632. First version December 2015; inspected version 12 dated 6 February 2018. The specific discussion about not using Dirac notation is in v12, section 5, p. 17. Sources: https://arxiv.org/abs/1512.06632, https://arxiv.org/pdf/1512.06632v1, https://arxiv.org/pdf/1512.06632v12.

**[S6] Dubois, Francois, and Zeno Toffano (2016).** *Eigenlogic: A quantum view for multiple-valued and fuzzy systems*, arXiv:1607.03509. Primary paper inspected at the operator/projector and Dirac notation passages. Source: https://arxiv.org/abs/1607.03509.

**[S7] Cheng, Daizhan, and Hongsheng Qi (2010).** *A linear representation of dynamics of Boolean networks*, IEEE Transactions on Automatic Control 55(10), 2251-2258. Author PDF inspected for the representation and publication metadata. Source: https://lsc.amss.ac.cn/~hsqi/papers/TAC2010-LinearRepresentation.pdf.

**[S8] Mishchenko, Alan, Satrajit Chatterjee and Robert K. Brayton (2006).** *DAG-aware AIG rewriting: A fresh look at combinational logic synthesis*, DAC, 532-536, DOI 10.1145/1146909.1147048. Proceedings paper/record inspected for NPN/cut-reuse context and metadata. Source: https://web.cecs.pdx.edu/~mperkows/CLASS_573/febr-2007/p532-mishchenko.pdf.

**[S9] Zhou, Xuegong, Lingli Wang and Alan Mishchenko (2019).** *Fast adjustable NPN classification*

*using generalized symmetries*, ACM TRETS 12(2), Article 7, 16 pp., DOI 10.1145/3313917. Author PDF inspected. Source: https://people.eecs.berkeley.edu/~alanmi/publications/2019/trets19_npn.pdf.

**[S10] Carlet, Claude.** *Boolean Functions for Cryptography and Error Correcting Codes*, author-hosted chapter. Relevant inspection: Boolean functions, ANF and Mobius-transform discussion, especially section 2.1. Source: https://www.math.univ-paris13.fr/~carlet/chap-fcts-Bool-corr.pdf.

**[S11] Burris, Stanley, and H. P. Sankappanavar.** *A Course in Universal Algebra*, original 1981, corrected author-hosted edition 2012. General algebraic background and edition record; not a claimed exact antecedent printing the LM theorem. Source: https://www.math.uwaterloo.ca/~snburris/htdocs/UALG/univ-algebra2012.pdf.

**[S12] Post, Emil L. (1941).** *The Two-Valued Iterative Systems of Mathematical Logic*, Annals of Mathematics Studies 5. Publisher preview checked for the original record and substitution-system context. Source: https://api.pageplace.de/preview/DT0400.9781400882366_A27761476/preview-9781400882366_A27761476.pdf.

**[S13] Edwards, C. R. (1972).** *The Logic of Boolean Matrices*, The Computer Journal 15(3), 247-253, DOI 10.1093/comjnl/15.3.247. Publisher metadata/abstract, not a complete theorem-level reading. Source: https://academic.oup.com/comjnl/article-abstract/15/3/247/480639.

**[S14] Stern, August (1988).** *Matrix Logic*, North-Holland; the manuscript uses the extended title *Matrix Logic: Theory and Applications*. Publisher/book preview and catalog context checked; full monograph unavailable for this audit. Source: https://api.pageplace.de/preview/DT0400.9781483295497_A23889179/preview-9781483295497_A23889179.pdf.

**[S15] Mizraji, Eduardo (1992).** The manuscript lists *Vector logics: The matrix–vector representation of logical calculus*, Fuzzy Sets and Systems 50(2), 179-185. This audit did not complete independent primary-record verification of all those details or inspect the complete article. This is an explicit unresolved bibliography item, not a verified external citation.

**[S16] Mizraji, Eduardo (2008).** *Vector logic: A natural algebraic representation of the fundamental logical gates*, Journal of Logic and Computation 18(1), 97-121, DOI 10.1093/logcom/exm057. Publisher issue metadata and author-manuscript content used for vector/operator context. Source: https://academic.oup.com/logcom/issue/18/1.

**[S17] Droncheff, Brian (2018).** *Correspondence Matrices; Algorithms for Propositional Logic*, ResearchGate preprint, DOI 10.13140/RG.2.2.28036.37764. Primary record shows May labeling and an October 1 upload. Historical mathematical material was used selectively, not certified wholesale. Source: https://www.researchgate.net/publication/327980262_Correspondence_Matrices_Algorithms_for_Propositional_Logic.

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